Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which compound acts as an oxidising as well as reducing agent

(A) SO_2

(B) H_2SO_4

(C) Al_2O_3

(D) $K_2Cr_2O_7$

Q.2

Sulphuric acid reacts with PCl_5 to give

(A) Thionyl chloride

(B) Sulphur monochloride

(C) Sulphuryl chloride

(D) Sulphur tetrachloride

Q.3

When sulphur is boiled with Na_2SO_3 solution, the compound formed is

(A) Sodium sulphide

(B) Sodium sulphate

(C) Sodium persulphate

(D) Sodium thiosulphate

Q.4

Which of the following after burning at room temperature gives gaseous oxide

(A) H

(B) Na

(C) S

(D) He

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Q.5

The deep blue colour produced on adding excess of ammonia to copper sulphate is due to presence of

(A) Cu^{2+}

(B) $[Cu(NH_3)_4]^{2+}$

(C) $[Cu(NH_3)_6]^{2+}$

(D) $[Cu(NH_3)_2]^{2+}$

Q.6

From the following information $X + H_2SO_4 \rightarrow Y$ (a colourless and irritating gas)
$Y + K_2Cr_2O_7 + H_2SO_4 \rightarrow$ (green coloured solution)
Identify the pair X and Y.

(A) $\overline{Cr}, HCl$

(B) SO_3^{2-}, SO_2

(C) S^{2-}, H_2S

(D) CO_3^{2-}, CO_2

Q.7

H_2SO_4 and H_2SO_3 can be distinguished by the addition of

(A) magnesium powder

(B) $NaHSO_4$ solution

(C) $FeCl_3$ solution

(D) litmus solution

Q.8

Which of the following is the key step in the manufacture of sulphuric acid?

(A) Burning of sulphur or sulphide ores in air to generate SO_2.

(B) Conversion of SO_2 to SO_3 by the reaction with oxygen in presence of catalyst.

(C) Absorption of SO_3 in H_2SO_4 to give oleum.

(D) Both (b) and (c).

Q.9

$FeCl_3$ solution on reaction with SO_2 changes to

(A) $FeCl_2$

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(B) $Fe_2(SO_4)_3$

(C) $Fe_2(SO_3)_3$

(D) $FeSO_4$

Q.10

Hydrolysis of one mole of peroxodisulphuric acid
produces

(A) two moles of sulphuric acid.

(B) two moles of peroxomonosulphuric acid.

(C) one mole of sulphuric acid and one mole of peroxomonosulphuric acid.

(D) one mole of sulphuric acid, one mole of peroxomonosulphuric acid and one mole of hydrogen peroxide.

Q.11

Concentrated H_2SO_4 is not used to prepare HBr from KBr because it

(A) oxidizes HBr.

(B) reduces HBr.

(C) causes disproportionation of HBr.

(D) reacts too slowly with KBr.

Q.12

Which reaction represents the oxidizing behaviour of H_2SO_4 ?

(A) $2PCl_5 + H_2SO_4 \rightarrow 2POCl_3 + 2HCl + SO_2Cl_2$

(B) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

(C) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

(D) $2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

Q.13

The number of $S-S$ bonds in SO_3, $S_2O_3^{2-}$
$S_2O_6^{2-}$ and $S_2O_8^{2-}$ respectively are

(A) $1, 0, 0, 1$

(B) $1, 0, 1, 0$

(C) $0, 1, 1, 0$

(D) $0, 1, 0, 1$

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Q.14

Iron sulphide is heated in air to form A, an oxide of sulphur. A is dissolved in water to give an acid. The basicity of this acid is

(A) 2

(B) 3

(C) 1

(D) zero

Q.15

Which of the following is incorrect?

(A) M.p of monoclinic sulphur $>$ m.p. of rhombic sulphur.

(B) Specific gravity of rhombic sulphur $>$ specific gravity of monoclinic sulphur.

(C) Monoclinic sulphur is stable below $369\ \mathrm{K}$.

(D) Both rhombic sulphur and monoclinic sulphur have $\mathrm{S_8}$ molecules.

Q.16

Which of the following form of the sulphur shows paramagnetic behaviour?

(A) $\mathrm{S_8}$

(B) $\mathrm{S_6}$

(C) $\mathrm{S_2}$

(D) All of these

Q.17

Which of the following has $\mathrm{p}\pi - \mathrm{d}\pi$ bonding ?

(A) $\mathrm{NO_3^-}$

(B) $\mathrm{SO_3^{2-}}$

(C) $\mathrm{BO_3^{3-}}$

(D) $\mathrm{CO_3^{2-}}$

Q.18

By which of the following $\mathrm{SO_2}$ is formed?

(A) Reaction of dil. $\mathrm{H_2SO_4}$ with $\mathrm{O_2}$

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(B) Hydrolysis of dil. H_2SO_4

(C) Reaction of conc. H_2SO_4 with Cu

(D) None of these

Q.19

The acid which has a peroxy linkage is

(A) Sulphurous acid

(B) Pyrosulphuric acid

(C) Dithionic acid

(D) Caro's acid

Q.20

'O - O' peroxy bond is present in :-

(A) H_2SO_5

(B) HNO_4

(C) $H_2S_2O_8$

(D) All of these

Answers & Solutions

Q.1 Answer:

SO_2

Solution:

The minimum and maximum oxidation number of S are -2 and $+6$ respectively. Since the oxidation number of S in SO_2 is $+4$, therefore it can be either increased or decreased. Therefore SO_2 behaves both as an oxidising as well as reducing agent.

Q.2 Answer:

Sulphuryl chloride

Solution:

$$HO-SO_2-OH+PCl_5 \rightarrow Cl-SO_2-OH+POCl_3+HC$$
$$HO-SO_2-OH+2PCl_3 \rightarrow Cl-SO_2-C+2EOCl_3+2HCl$$

Q.3 Answer:

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Sodium thiosulphate

Solution:

$$Na_2SO_3 + S \rightarrow Na_2S_2O_3$$

Q.4 Answer:

S

Solution:

$$S + O_2 \rightarrow SO_2$$

Q.5 Answer:

$[Cu(NH_3)_4]^{2+}$

Solution:

$$CuSO_4 + 4NH_3 \rightarrow [Cu(NH_3)_4]SO_4$$

Blue complex due to $Cu(NH_3)_4^{2+}$

Q.6 Answer:

SO_3^{2-}, SO_2

Solution:

$$\underset{(X)}{SO_3^{2-}} + 2H^{\oplus} \rightarrow SO_2 + H_2O$$

$$\underset{(Y)}{3SO_2} + K_2Cr_2O_7 + H_2SO_4 \rightarrow$$

$$K_2SO_4 + Cr_2(SO_4)_2 + H_2O$$

Green coloured solution

Q.7 Answer:

$FeCl_3$ solution

Solution:

$FeCl_3$ acts as oxidant and H_2SO_3 as reductant

Q.8 Answer:

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Conversion of SO_2 to SO_3 by the reaction with oxygen in presence of catalyst.

Solution:

The key step in the manufacture of H_2SO_4 is catalytic oxidation of SO_2 with O_2 to give SO_3 in presence of V_2O_5.

Q.9 Answer:

$FeCl_2$

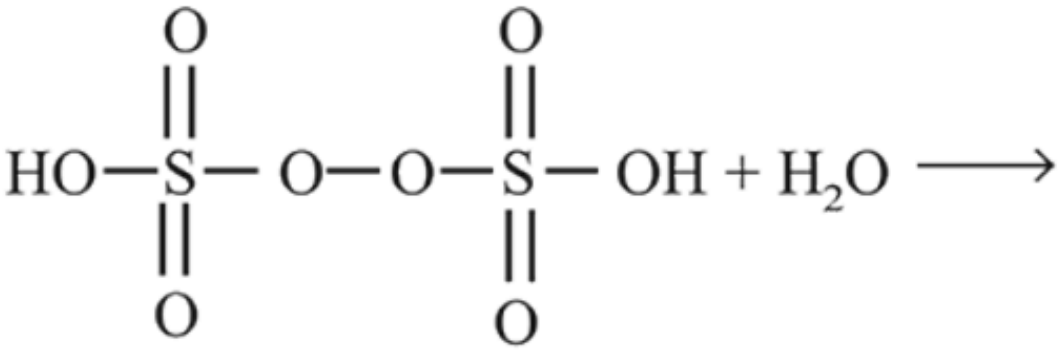
Solution:

$$2FeCl_3 + SO_2 + 2H_2O \rightarrow 2FeCl_2 + H_2SO_4 + 2HCl$$

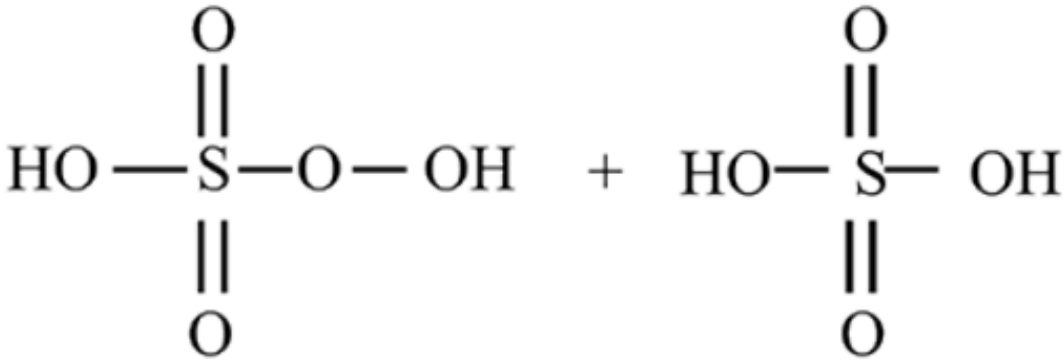
Q.10 Answer:

one mole of sulphuric acid and one mole of peroxomonosulphuric acid.

Solution:



peroxodisulphuric acid



peroxomonosulphuric acid sulphuric acid

Q.11 Answer:

oxidizes HBr.

Solution:

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$2HBr + H_2SO_4 \rightarrow Br_2 + 2H_2O + SO_2$

HBr is oxidised to Br_2

Q.12 Answer:

$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$

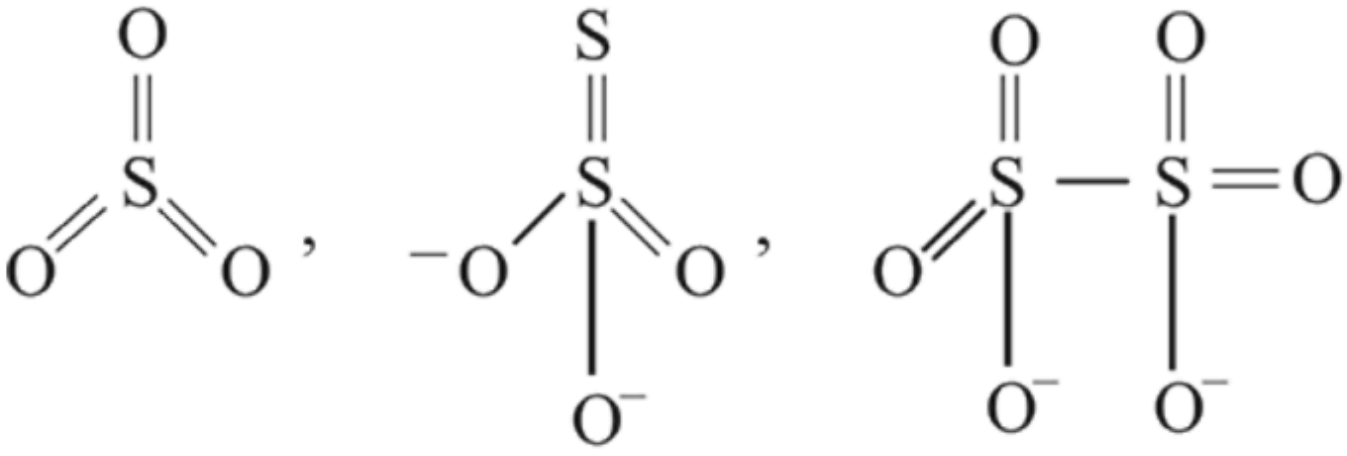
Solution:

$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$ **(HI is oxidised to I_2) (H_2SO_4 is reduced to SO_2)**

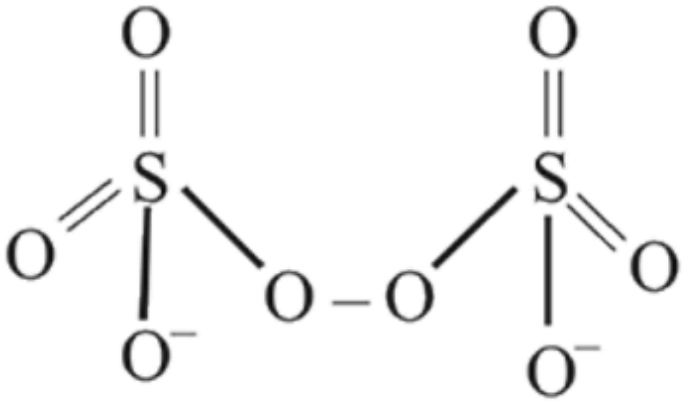
Q.13 Answer:

$0, 1, 1, 0$

Solution:



(SO_3) $(S_2O_3^{2-})$ $(S_2O_6^{2-})$



$(S_2O_8^{2-})$

Hence (c) is the correct option.

Q.14 Answer:

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Solution:

$$4FeS + 7O_2 \rightarrow 2Fe_2O_3 + 4SO_2$$
$$SO_2 + H_2O \rightarrow H_2SO_3 \rightleftharpoons 2H^+ + SO_3^{2-}$$

It is dibasic

Q.15 Answer:

Monoclinic sulphur is stable below $369\ K$.

Solution:

Monoclinic sulphur is stable above $369\ K$.

Q.16 Answer:

S_2

Solution:

S_2 is paramagnetic. It contains two unpaired electrons in the antibonding π^* orbital

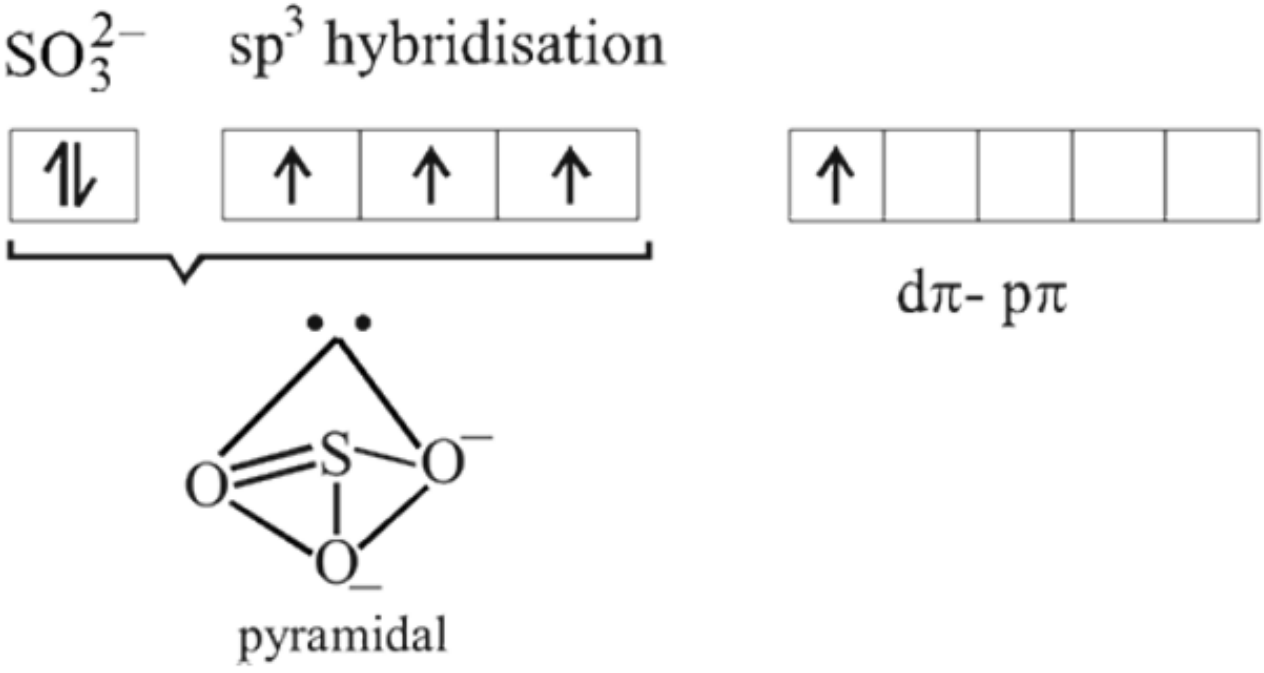
Q.17 Answer:

SO_3^{2-}

Solution:

$p\pi - d\pi$ bonding is present in SO_3^{2-}, N, B,

C have no vacant d atomic arbitals.



Q.18 Answer:

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Reaction of conc. H_2SO_4 with Cu

Solution:

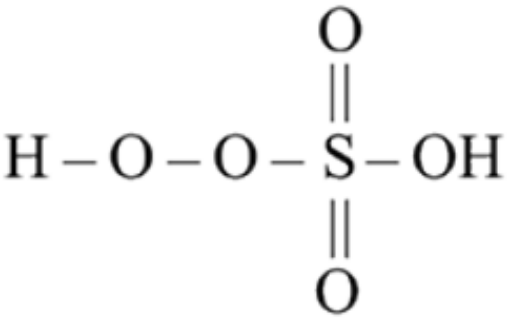
$$Cu + 2H_2SO_4(conc) \longrightarrow CuSO_4 + SO_2 + 2H_2O$$

Q.19 Answer:

Caro's acid

Solution:

Caro's acid is H_2SO_5 which contains one S – O – O – H peroxy linkage. It is also known as permonosulphuric acids.

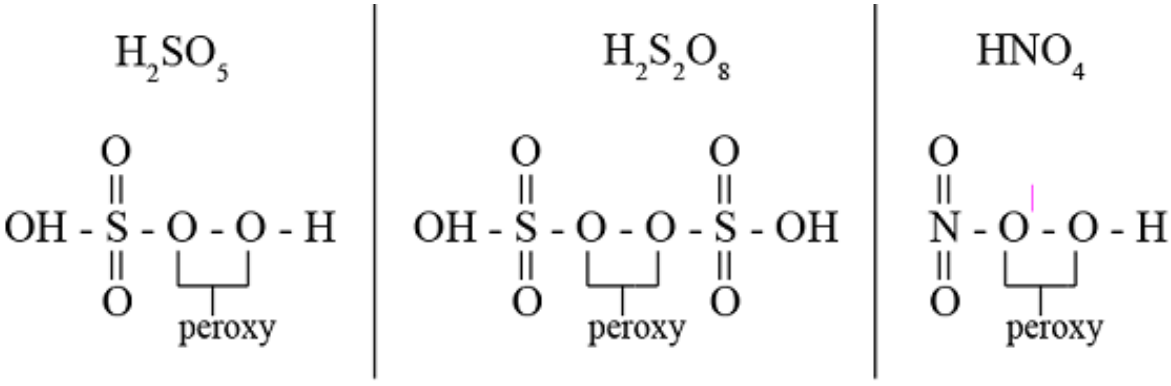


Caro's acid

Q.20 Answer:

All of these

Solution:



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